

# DR. LIAM MOROY

Researcher in Stochastic Modelling, Monte Carlo Methods & Machine Learning.

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## SUMMARY

PhD researcher specialising in **stochastic differential equations**, **Bayesian inference** and **large-scale Monte Carlo simulation**, currently a Research Associate at Heriot-Watt University. Four years designing and calibrating stochastic sampling algorithms for high-dimensional, ill-posed inference problems, in production-grade Python/PyTorch on HPC clusters. Independently applying this toolkit to quantitative finance: a score-based generative model of **S&P 500 implied volatility surfaces** that learns no-arbitrage structure directly from 13 years of options data.

## SELECTED PROJECTS

### Generative Modelling of Implied Volatility Surfaces

[github.com/LiamMry/iv-surface-diffusion](https://github.com/LiamMry/iv-surface-diffusion) — independent project, 2026

- Score-based diffusion model (VP-SDE, attention U-Net) trained on 3,500 daily SPY option surfaces (2010–2023), generating realistic implied-volatility surfaces from noise.
- With no explicit no-arbitrage constraint, generated surfaces learn financial consistency from data: calendar-spread and butterfly-spread violation rates approach the market baseline.
- Unconditional and conditional sampling pipelines; systematic comparison of ODE/SDE samplers (Euler–Maruyama, Heun, DPM-Solver) trading off accuracy against function-evaluation budget.

Stochastic simulation    Options data    No-arbitrage diagnostics    PyTorch

### Super-Calibration of Guided Stochastic Samplers

Research project — paper in preparation, 2025–2026

- Framework for calibrating the guidance strength of diffusion-based posterior samplers so that sampled marginals match a target distribution family — closed-form calibration weights, on-policy sequential (“backward”) calibration correcting co-variate shift.
- Analysis of weak vs. strong convergence trade-offs: why  $L^2$ -optimal calibration of a stochastic sampler can degrade distributional (weak-sense) accuracy — the notion of accuracy relevant to expectation estimates.

Calibration    SDE discretisation    Weak/strong convergence    Kernel methods

## PROFESSIONAL EXPERIENCE

### Research Associate — Generative Models & Bayesian Inverse Problems

Heriot-Watt University (BISC)

📅 Dec. 2025 - Present    📍 Edinburgh, Scotland

- Research on stochastic (diffusion-based) sampling algorithms as learned priors for high-dimensional Bayesian inverse problems, with uncertainty quantification over reconstructions.
- Applications to computed tomography for nuclear, medical and non-conventional optical imaging: recovering signals from severely limited, noisy measurements.

Python    PyTorch/CUDA    SDE samplers    Bayesian inversion    HPC/Slurm

### R&D Internship — Deep learning for tomographic reconstruction of compressible flows

ONERA (DTIS/MIC)

📅 Feb. 2022 - Aug. 2022    📍 Paris, France

- Designed and evaluated learned prior models for Bayesian reconstruction from limited measurements.

Python    PyTorch/CUDA    Generative AI    Bayesian inversion

## R&D Internship — Deep learning for nebula removal in astronomical images

Astrophysics Laboratory of Bordeaux (LAB) & IMS Laboratory

📅 Jul. 2021 - Sep. 2021

📍 Bordeaux, France

- Trained regressors from noisy ground truths; transfer learning from synthetic datasets.

Python

PyTorch/CUDA

C/C++

Image processing

## EDUCATION

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### Ph.D. — Université Paris-Saclay / ONERA (DTIS/MIC)

📅 2022 - 2025

📍 Paris, France

- Stochastic (diffusion-based) generative models for high-dimensional Bayesian inverse problems.
- Core mathematical toolkit: Itô calculus, SDEs and their discretisation, change of measure (Girsanov), Monte Carlo methods, score-based sampling, convergence analysis.
- Supervised by Prof. Jean-François Giovannelli, Dr. Frédéric Champagnat, Dr. Guillaume Bourmaud.

### MSc — Complex Systems Engineering

University of Bordeaux, concurrent with the engineering degree below

📅 2022

📍 Bordeaux, France

### BSc + MSc — Electronics Engineering, specialised in Signal Processing

ENSEIRB-MATMECA engineering school, Bordeaux INP

📅 2017 - 2022

📍 Bordeaux, France

## PUBLICATIONS & TALKS

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### Publications:

- “On the Posterior Gap in Plug&Play Diffusion Methods for Sparse-View CT”, IEEE JSTSP 2026 Special Issue on High-Dimensional Imaging, [L. Moroy](#), G. Bourmaud, F. Champagnat, J.-F. Giovannelli.
- “Evaluating the Posterior Sampling Ability of Plug&Play Diffusion Methods in Sparse-View CT”, ICASSP 2025, [L. Moroy](#), G. Bourmaud, F. Champagnat, J.-F. Giovannelli.

### Selected Talks:

- “Improving the Posterior Sampling of Plug&Play Diffusion Models by Super-Calibration: application to Poisson inverse problems”, 7th SLSIP Workshop, May 2026, Krujë, Albania.
- “Plug&Play Diffusion Models for General Imaging Inverse Problems”, ISSS Seminar, Mar. 2026, Heriot-Watt University, Edinburgh, UK.
- “Plug-and-Play Diffusion Methods for Posterior Sampling in Tomographic Reconstruction”, AI in IMS, Jan. 2025, Bordeaux.
- “Evaluating the Posterior Sampling Ability of Plug&Play Diffusion Methods in Sparse-View CT”, GdR IASIS, Dec. 2024, Henri Poincaré Institute, Paris.

## TEACHING

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- **University of Bordeaux** — MSc TA, Introduction to noise and random signals (2023/24, 2024/25).
- **ENSEIRB-MATMECA** — MSc TA, Generative modelling with diffusion models (2024/25); Freshman TA, Introduction to programming in C/C++ (2023/24, 2024/25).

## SKILLS

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### Mathematics

- Stochastic calculus (Itô, Girsanov), SDEs & discretisation schemes
- Monte Carlo methods, variance reduction, convergence analysis
- Bayesian inference, inverse problems, uncertainty quantification
- Statistical / machine learning, kernel methods

### Programming

- Python (advanced), C (advanced), C++ (intermediate)
- PyTorch / CUDA, NumPy, pandas

### Tools & Workflow

- Git / GitHub,  $\LaTeX$
- Linux / Bash, Slurm / HPC clusters

### Languages

- French (native)
- English (TOEIC 950 – fluent)
- Spanish (B1 – intermediate)
- Japanese (beginner)